

Hero Deck Dominance and Community Sentiment in Clash Royale

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Abstract

This paper examines the competitive and monetization implications of the hero mechanic introduced in Clash Royale in December 2025, analyzing 4.5 million top 1,000 ladder games across five competitive seasons from December 2025 through May 2026. Statistical analysis comparing hero and non-hero deck win rates at both population and elite levels was complemented by VADER sentiment analysis of 3,499 tweets collected from 11 active competitive players on Twitter. Results indicate hero decks were statistically dominant over non-hero decks in four of five seasons, with advantages more pronounced among elite top 100 players than across the broader top 1,000 population, peaking at a 95% hero adoption rate among top 100 finishers in Season 4. Sentiment analysis revealed that community perception did not track hero dominance in a simple linear way, with the most statistically dominant season paradoxically corresponding to the most positive player sentiment: a divergence attributed to concurrent improvements in hero accessibility. Together these findings suggest that competitive advantage and perceived fairness represent distinct dimensions of mobile game monetization, with implications for how developers balance content monetization and player satisfaction.

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1 Introduction

Mobile gaming has always been an outlier in comparison to other gaming setups. While generally more limiting in mechanics, performance, and graphical fidelity, mobile gaming compensates with its high accessibility, convenience, and relatively low installation costs. However, maintaining such games requires sustainable monetization, presenting developers with a fundamental dilemma: will they center monetization on progression systems, in-game mechanics, or cosmetic accessories? How much of the game can be monetized before commercialization begins to hinder gameplay and player retention? Mobile game developers must find a balance that simultaneously maximizes player satisfaction and revenue, a balance that, when mishandled, can fracture the relationship between a game and its community.

Clash Royale, developed by Supercell and released in 2016 [1], is one of the most enduring and commercially successful mobile games globally. The game fuses real-time player versus player combat with strategic card collection, where players build eight-card decks from a roster of collectible cards obtained and upgraded using in-game currency. Gameplay introduces new players gradually through an arena progression system, unlocking cards with increasingly unique capabilities before culminating in a hyper-competitive ranked ladder where players compete for global standings. This competitive ecosystem, combined with the game’s ongoing monetization model, creates a unique environment for studying the relationship between spending and competitive advantage.

While most cards rely solely on player input for placement, Supercell introduced champions in 2021, cards with activatable special abilities triggered by the player at any point during the card’s lifetime for an elixir cost. Champions were designed to be slightly weaker than standard cards, compensating through their unique abilities. While initial balancing of new champions was often inconsistent, many players came to view them as a meaningful addition to skill expression after Supercell recalibrated the evenness between the strength of champion abilities and the weakness of champion base cards.

Four years later in 2025, Supercell introduced heroes: an adaptation of the champion mechanic

applied to existing cards in the roster. Unlike champions, which were obtainable through standard gameplay progression, heroes were largely inaccessible without significant luck from a fixed amount of loot boxes, or direct purchase through microtransactions tied to hero releases. Moreover, heroes were not balanced equivalently to the now stable champions as they provided absolute statistical buffs to already established cards, making hero forms meaningfully stronger than their base versions without requiring ability activation. This accessibility gap and mechanical advantage distinguishes heroes from prior content additions and positions them as a particularly consequential case study in mobile game monetization.

This paper addresses two related questions: First, do hero-containing decks demonstrate statistically significant competitive advantages over non-hero decks at top ladder? Second, how did competitive players express sentiment throughout the hero era, and did community frustration align with periods of statistically measured hero dominance?

Using 4.5 million top ladder games across five competitive seasons from December 2025 through May 2026, combined with end-of-season deck data from the top 100 global finishers and sentiment analysis of active competitive players on Twitter, this paper finds that hero-containing decks demonstrate statistically significant win rate advantages over non-hero decks in four of five seasons examined, with competitive advantages more pronounced among elite players than across the broader top ladder population. Complementary sentiment analysis reveals that community frustration does not linearly track hero dominance, with the most statistically dominant season paradoxically corresponding to relatively positive sentiment. This misalignment perhaps suggests a disconnect between quantitative competitive advantage and overall perceived balancing by top players. The remainder of this paper is structured as follows: Section 2 provides background on Clash Royale’s competitive ecosystem and hero mechanic. Section 3 describes the data sources and collection methodology. Section 4 details the statistical and sentiment analysis methodology. Section 5 presents statistical results. Section 6 presents sentiment findings. Section 7 discusses the implications of both components. Section 8 concludes with limitations and directions for future research, followed by references and an appendix.

2 Background

Clash Royale's progression system is built around trophies, a currency earned and lost through player versus player matches that serves as the primary measure of a player's standing in the game. Trophy Road introduces new players to the game's card pool progressively, with each 300 to 500 trophy threshold corresponding to a new arena where additional cards become available for collection. Beyond a season-determined threshold past 12,000 trophies, players unlock the game's ranked mode, which operates through a league system where victories advance the player one step forward and defeats push them one step back, with save points at lower league thresholds preventing excessive regression.

The highest tier of ranked play is Ultimate Champion, where the step system is replaced by a rating-based format analogous to trophy road but reserved for the game's most competitive players. Starting rating in Ultimate Champion is calculated using the player's win rate of the ranked games played before attaining Ultimate Champion, with gains and losses calibrated to reflect the competitiveness of the player pool at that level after a player gets assigned their starting rating. Finishing a season in the top 1,000 globally is widely regarded as a meaningful benchmark of competitive achievement, reflected in an exclusive profile badge awarded to players after the season ends. The top 100 represents the pinnacle of competitive ladder, where the game's best players converge to define the meta and demonstrate the highest level of mechanical and strategic skill the game has to offer.

End of season dynamics are particularly relevant to this study's methodology and represent the most consequential period of any competitive season in Clash Royale. In the final days of each season, ranked play becomes extremely competitive as players push for personal rating records and global leaderboard placements. A distinct feature of this period is the accelerated inflation of ranking thresholds where the rating required to maintain a top 1,000 or top 100 position rises rapidly as more players push simultaneously, meaning that even a player sitting comfortably at rank 50 with two days remaining cannot safely stop competing without risking losing their position entirely. This threshold inflation effectively forces elite players to remain actively engaged through

the final hours of the season, guaranteeing a concentration of the game's best players competing at maximum intensity in the last two days. Consequently, end of season games capture the highest quality of play observed across the entire season, with top players performing at their absolute peak against similarly motivated opponents under maximum competitive pressure. It is for this reason that this study restricts its primary battle data analysis to the last two days of each season which helps ensure findings reflect peak competitive play rather than the experimentation and variance characteristic of earlier season periods.

Clash Royale's card slot system underwent significant evolution throughout the hero era. Prior to March 16th, 2026, players had access to four special card slots: two designated for evolution forms and two for hero forms, allowing a maximum of four special cards to be deployed simultaneously. The March 16th update consolidated this system into three slots: one dedicated evolution slot, one dedicated hero slot, and one flexible slot that players could assign to either an evolution or hero form at their discretion. This change effectively reduced the maximum number of simultaneously deployable special cards from four to three, meaningfully altering deck construction dynamics mid-season.

Heroes were introduced in a staggered release schedule beginning December 1st, 2025, with Knight, Musketeer, Giant, and Mini PEKKA launching simultaneously as the first wave. Subsequent releases followed approximately monthly: Wizard and Ice Golem in January 2026, Goblins and Mega Minion in February, Magic Archer and Barbarian Barrel in March, Balloon in April, and Dark Prince and Bowler in May. This release cadence ensured a steady stream of new hero content throughout the study period, with each release representing both a gameplay addition and a monetization event.

The mechanism by which players could obtain heroes is central to understanding their monetization implications. Upon the hero system's launch, Supercell provided each player one free hero selected at random. Beyond this initial grant, heroes could only be unlocked through a shard-based system requiring 200 hero shards for a random hero unlock. Shards were obtainable exclusively through limited seasonal events and the paid Pass Royale, with drop rates distributed across tiered

loot boxes: 2-star boxes yielded 5 shards at a 34.3% drop rate, 3-star boxes yielded 10 shards at 52.1%, 4-star boxes yielded 40 shards at 12.6%, and 5-star boxes yielded 200 shards, an instant random hero unlock, at a rate of 1%. This drop structure meant that free to play players had extremely limited and probabilistically uncertain access to heroes in the system's initial months. Supercell revised the hero acquisition system twice during the study period. In February 2026, the randomized shard drop system through Pass Royale was replaced with a fixed shard allocation per season, providing more predictable but still insufficient progression as the fixed allocation represented less than 50% of the 200 shards required for a single unlock. The March 16th update further refined the system by allowing players to select their desired hero upon reaching the 200 shard threshold, replacing the previous random unlock mechanic. These changes suggest Supercell was responding to community feedback regarding hero accessibility, though the fundamental barrier of requiring sustained Pass Royale investment remained throughout the study period.

The introduction of heroes in December 2025 was met with widespread community skepticism and frustration. Players broadly perceived the initial hero releases as poorly balanced and game-breaking, with hero forms providing absolute statistical advantages that disrupted the status quo. This reception was not entirely surprising given the game's history since similar complaints had accompanied the introduction of evolutions in 2023 and the raising of the maximum card level to level 15, both of which were criticized as monetization-driven decisions that created competitive gaps between paying and non-paying players. The hero era contributed to growing community skepticism toward Supercell's monetization approach, with heroes perceived as an escalation of prior concerns rather than an isolated incident.

The most prominent community flashpoint of the hero era occurred in early February 2026, when popular content creator Jynxzi publicly announced his departure from Clash Royale. His decision was attributed to several converging factors: dissatisfaction with the hero mechanic and its competitive implications, frustration over the introduction of a level 16 maximum card level which was perceived as unnecessary progression inflation, and a public statement by Supercell's CEO crediting the game's recent success primarily to a celebrity collaboration rather than to the

content creator community that had sustained the game’s audience and led to the game’s growth in the past year. The CEO’s statement prompted a broader creator response, with several prominent figures boycotting a Supercell-sponsored content creator event that was subsequently postponed. Jynxzi’s departure and the surrounding controversy represented continued discontent within Clash Royale’s competitive community toward Supercell, providing important qualitative context for interpreting the quantitative patterns observed in this study.

Together, these dynamics position the hero era as a particularly instructive case study in mobile game monetization. The combination of mechanically controversial new content, restricted accessibility, level cap inflation, and a high profile community departure created conditions in which the relationship between competitive advantage and perceived fairness became unusually noticeable, making the December 2025 through May 2026 period an ideal window for empirical investigation of pay to win dynamics in a live competitive mobile game.

3 Data

3.1 Battle Data

The primary dataset consists of battle logs scraped daily from RoyaleAPI [2], capturing all games played within the top 1,000 global rankings in Clash Royale’s Ultimate Champion ladder. RoyaleAPI was chosen as the data source as the platform does not retain historical game logs indefinitely, necessitating daily collection to preserve a record of top ladder activity. Data collection was conducted by an independent collaborator using an automated scraping pipeline, covering the period from December 1st, 2025 through May 16th, 2026, which spanned the first five complete seasons of the hero era.

Each observation represents a single completed match between two players, containing the following variables: player tag and Ultimate Champion rating for both participants, full deck composition for each player including all eight playing cards and tower troop, card levels, an evolution level indicator denoting whether each card was played in standard, evolution, or hero form, and

crown count for each player enabling determination of match outcome. In total the dataset comprises approximately 4.5 million individual matches across 167 daily files, providing a comprehensive record of elite competitive play throughout the study period.

One limitation of the dataset is its coverage boundary since data collection ceased on May 16th, 2026, meaning the final two days of Season 6 fall outside the study window. As a result, Season 6 is excluded from the primary win rate analysis despite being included in the leaderboard analysis described in Section 3.2. Additionally, as data was sourced through a third party scraping pipeline rather than directly from the official Clash Royale API, minor data quality issues cannot be entirely ruled out, though no systematic errors were identified during processing.

3.2 Top 100 Leaderboard Data

End of season leaderboard data was manually compiled from RoyaleAPI’s seasonal blog posts [3], which publish infographics and player account links documenting the top 100 global finishers at the conclusion of each competitive season. For each of the six seasons spanning December 2025 through June 2026, the following variables were recorded for all 100 finishers: season identifier, finishing rank, player tag, deck name as labeled by RoyaleAPI, a binary indicator of whether the deck contained a hero form, the specific hero or heroes used where applicable, and the player’s finishing Ultimate Champion rating. The resulting dataset comprises 600 entries across six seasons.

This dataset contains three characteristics worth noting before proceeding to analysis. First, player tags from the leaderboard data were cross-referenced with the battle data to enable filtering of game logs to elite players specifically. Second, Season 6 was notably shorter than the other five seasons in the study period, resulting in deflated finishing ratings relative to other seasons, a structural difference acknowledged when interpreting cross-season comparisons. Third, as data was compiled manually from infographic images and player profile links, minor transcription errors cannot be entirely excluded, though the dataset was reviewed for consistency prior to analysis.

3.3 Sentiment Data

Sentiment data was collected from Twitter/X using Apify’s Advanced X Twitter Profile Scraper [4], targeting a curated list of 50 active competitive Clash Royale players identified through RoyaleAPI’s Twitter leaderboard and community knowledge of prominent ladder players. Due to platform rate limits and scraping constraints, complete timeline data was successfully retrieved for 11 of the 50 targeted accounts, covering the period from November 2025 through July 2026. Each observation represents a single tweet and includes the following variables: author handle, full tweet text, timestamp, language classification, and engagement metrics including likes, replies, reposts, and view count.

Raw data comprised 9,046 tweets across the collection period. Prior to sentiment analysis, three sequential filters were applied: a language filter retaining only English language tweets, a minimum character length filter removing tweets under 30 characters unlikely to contain meaningful sentiment signal, and a keyword filter retaining only tweets containing terms related to Clash Royale gameplay, balance, or monetization. After filtering, 3,499 tweets were retained for sentiment analysis, distributed across six seasons with between 276 and 861 tweets per season.

Several limitations of this dataset are acknowledged. First, the sample of 11 accounts represents a convenience sample rather than a systematic sample of competitive player opinion. However, the successfully scraped accounts were among the most actively posting players in the targeted list, collectively producing 9,046 raw tweets at an average of approximately 822 tweets per account, providing sufficient tweet density per season for stable sentiment aggregation. Results should nonetheless be interpreted as directional observations rather than generalizable community sentiment. Second, competitive gaming communities frequently communicate frustration through video clips of broken interactions accompanied by minimal caption text: a communication style that text-based sentiment analysis cannot capture, likely resulting in systematic underestimation of negative sentiment in seasons where controversial interactions were prevalent. Third, VADER’s known limitations with sarcasm and gaming community vernacular introduce additional measurement uncertainty. These limitations are discussed further in Section 8.

4 Methodology

4.1 Battle Data Processing

All data processing and analysis was conducted in Python, primarily utilizing the pandas library for data manipulation and scipy for statistical testing. The following subsections describe the analytical tools used for each component of the study.

Daily CSV files spanning the study period were loaded via Python and combined into a single dataframe comprising approximately 4.5 million observations. Dates were parsed directly from filenames rather than timestamps. Each observation was assigned to a competitive season based on predetermined season start and end dates sourced from RoyaleAPI's seasonal records. The dataset was then filtered to retain only games played within the final two days of each season, yielding 421,436 observations across five seasons for the primary analysis.

4.2 Hero Deck Detection

Hero deck identification relied on two criteria applied simultaneously: card identity and evolution level indicator. A deck was classified as containing a hero if any of the fourteen hero-eligible card IDs appeared in the deck with an evolution level value of two, indicating active hero form deployment as opposed to standard form (zero) or evolution form (one). Hero eligibility was further constrained by release date, and a card was only flagged as a hero if its hero form had been released prior to the season end date, preventing misclassification in seasons predating a given hero's introduction. This detection logic was applied independently to both player one and player two in each game record, producing binary hero deck indicators for each side of every match.

4.3 Win Rate Analysis and Statistical Testing

Match outcomes were determined by comparing crown counts between the two players, with the player recording the higher crown count designated as the winner. For each season, games

were classified into four matchup categories: hero versus hero, hero versus non-hero, non-hero versus hero, and non-hero versus non-hero. Win rate analysis focused exclusively on hero versus non-hero matchups to produce a clean comparison of hero and non-hero deck performance under equivalent competitive conditions. Hero win rate was computed as the proportion of hero versus non-hero matchups won by the hero deck across both player positions.

Statistical significance was assessed using two-sample independent t-tests comparing binary win outcomes between hero and non-hero decks within each season, with a significance threshold of $p < 0.05$. While this population-level analysis provides useful descriptive evidence of hero advantage across the broader competitive ladder, it is subject to an important confound: top 1,000 players span a wide range of skill levels and many are constrained by individual skill ceilings rather than deck strength. A player stuck at a particular rating band will produce win rates reflecting their personal ceiling rather than the true potential of their deck, introducing noise that may obscure the mechanical advantage conferred by heroes. This population-level analysis is therefore treated as preliminary and descriptive, establishing broad patterns rather than definitive evidence of hero mechanical advantage.

4.4 Elite Player Filtering

The primary and most rigorous analysis of this study focuses on games involving top 100 seasonal finishers. These are players who have demonstrably extracted the maximum competitive potential from their chosen deck by virtue of achieving the highest global rankings. Unlike the broader top 1,000 population, top 100 finishers cannot be said to be held back by skill limitations; their finishing rank is direct evidence that they have pushed their deck to its ceiling. By filtering to games involving these players, the analysis isolates deck strength from player skill as cleanly as the available data permits.

Player tags from the manually compiled leaderboard dataset were cross-referenced against player tag columns in the battle data, retaining any game in which either participant appeared in the top 100 finisher list for that season. This filtering yielded 143,568 observations across five

seasons. The same win rate computation and t-test procedure described in Section 4.3 was applied to this elite subset, with results interpreted as the primary evidence of hero mechanical advantage at peak competitive play.

4.5 Sentiment Analysis Pipeline

Tweet text was scored using VADER [5], a rule-based sentiment analysis tool designed specifically for social media text. VADER produces a compound sentiment score ranging from negative one to positive one for each tweet, where values below negative 0.05 are classified as negative, values above 0.05 as positive, and values in between as neutral. To account for systematic differences in baseline negativity across accounts, each tweet’s raw VADER score was normalized by subtracting the posting account’s mean sentiment score across the full study period. This normalization ensures that seasonal sentiment comparisons reflect deviations from each player’s individual baseline rather than absolute sentiment levels.

Normalized sentiment scores were aggregated to the season level by computing mean values across all retained tweets within each season. Complementary keyword frequency analysis was conducted by counting occurrences of terms associated with pay to win complaints, gameplay frustration, and player departure across three predefined keyword categories per season. Together, normalized sentiment scores and keyword frequencies constitute the seasonal sentiment indices used in the comparative analysis presented in Section 6.

5 Statistical Results

When analyzing the relative strength of specific in-game mechanics, win rate is a common metric utilized to quantify certain advantages or disadvantages observed. Broadly overviewing the 421,436 total games played in the top 1,000, decks composed of heroes had a clear advantage over decks lacking them, with hero decks outperforming non-hero decks in 4 of the 5 seasons recorded. As shown in Figure 1, hero decks were observed to have a 0.77% edge over non-hero decks in

Season 1, which rapidly increased, reaching a peak of 2.74% by Season 4, before declining in Season 5.

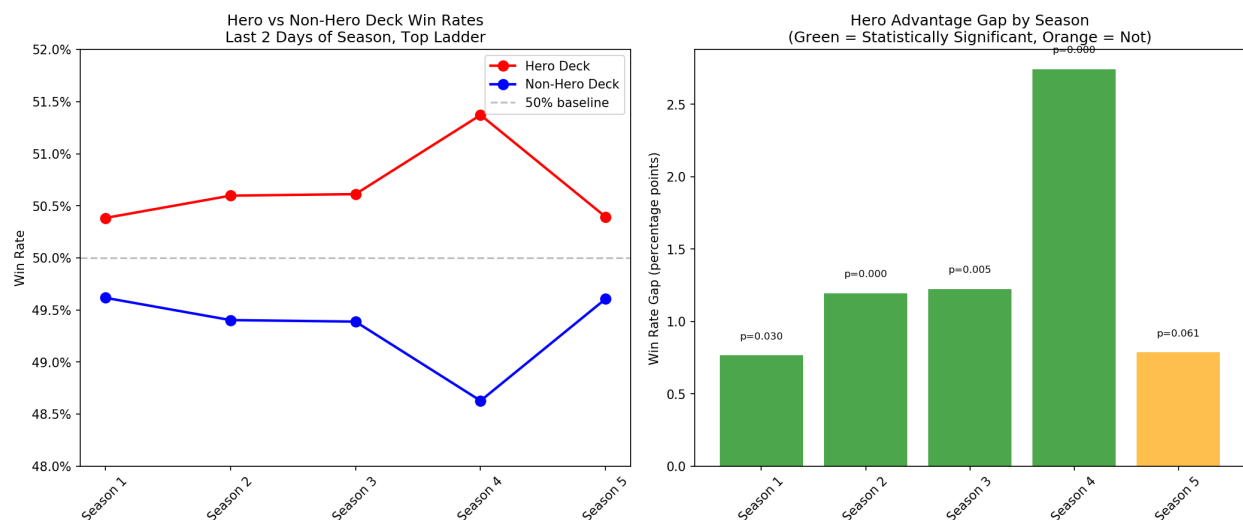


Figure 1: Hero vs Non-Hero Deck Win Rates and Advantage Gap by Season

Notably, individual hero win rate analysis revealed that newly released heroes consistently recorded the highest win rates in their debut seasons, with Magic Archer emerging as the strongest individual hero in Season 4 at a 50.66% win rate, suggesting that new hero releases drove seasonal spikes in overall hero advantage. While players were still unfamiliar to the hero meta in Season 1, the observed difference in win rate was found to be statistically significant, with a p-value of 0.03, well below the chosen significance threshold of 0.05 — a gap that continued to expand in the next 3 seasons. However, the statistically insignificant p-value of 0.061 observed in Season 5 indicates a waning of hero dominance, a finding consistent across both population and elite analyses and discussed further in Section 7. It should be noted that the t-tests reported here treat each game as an independent observation, though individual players contribute multiple games within the two-day analysis window. This non-independence likely understates the true standard errors, meaning the reported p-values should be interpreted as a lower bound on the true p-values rather than exact measures.

In addition to win rates, usage rates are also used to identify competitive advantages, as the dynamic environment of top ladder often forces players to adjust to an established meta. In accordance with the trends demonstrated through hero win rates, Figure 2 shows a reliance of top players on heroes, as usage rates were consistently above the 50% threshold, spiking in Season 4 at a considerable 85.64%, slightly decreasing to 80.65% in Season 5, where the difference of win rates was actually observed to be statistically insignificant.

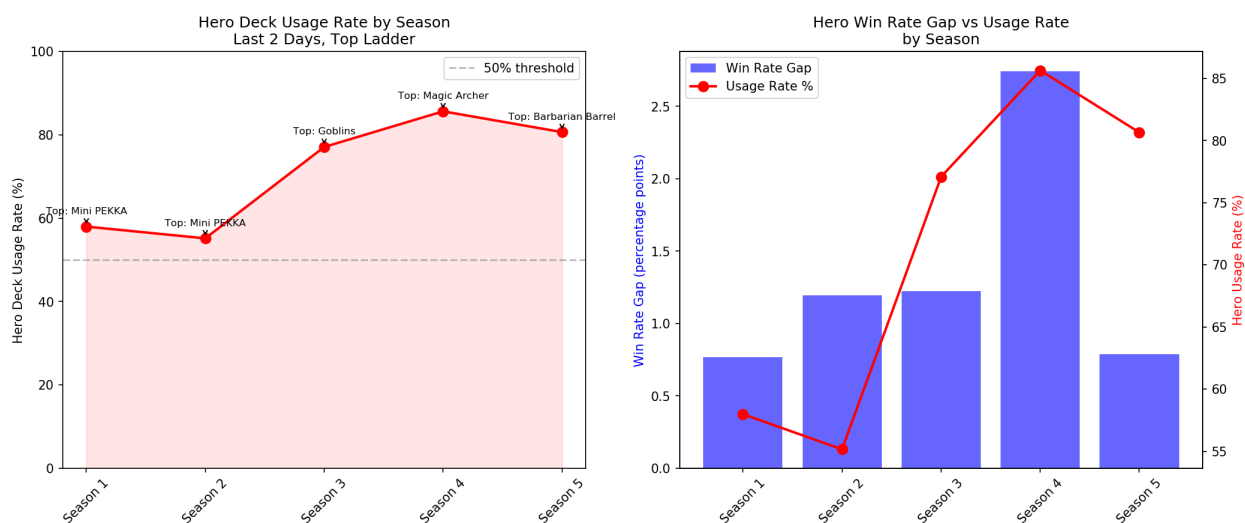


Figure 2: Hero Deck Usage Rate and Win Rate Gap vs Usage Rate by Season

While these population level findings establish broad patterns, top 1,000 players vary considerably in skill level and many are constrained by individual skill ceilings rather than deck strength, potentially skewing win rates. Therefore, a more rigorous analysis restricted to elite players would provide more insightful evidence, which is presented in the following subsection.

The trends observed in hero usage rates remain largely unimpacted by the narrowing of the scope, as a similar well-defined meta remains consistent between the top 1,000 and top 100. As observed in the second graph of Figure 3, hero adoption rates in the top 100 began at 60% in Season 1, when players were still experimenting with the newly added hero mechanic, rising to a remarkable 95% in Season 4.

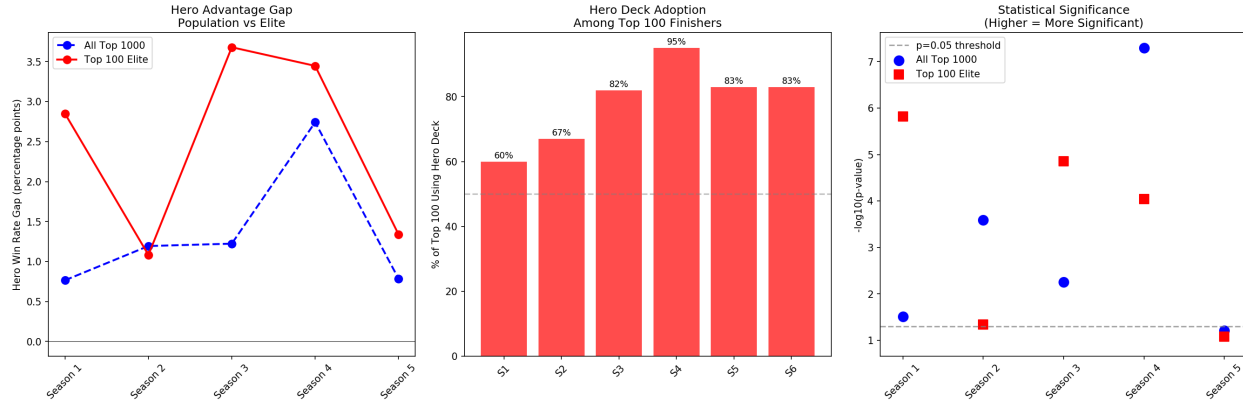


Figure 3: Hero Advantage Gap Population vs Elite, Hero Deck Adoption, and Statistical Significance

Contrary to usage rates, a notable difference emerges when comparing the win rate gaps between hero decks and non-hero decks of the top 1,000 versus the top 100. Coupling the statistics shown in the first and third graphs of Figure 3, hero advantage in the top 100 exceeded hero advantage in the top 1,000 in every single season, except for Season 2, where the measured differences were about the same, though the third graph indicates a rather considerable contrast in significance between the two. Season 2 represents the sole exception to this pattern, where the elite win rate gap of 1.08% marginally underperformed the population gap of 1.19%. This may reflect the early stage of hero adoption among elite players. With Wizard and Ice Golem releasing mid-season in January 2026, top players were still experimenting with newly available hero forms rather than running fully optimized hero strategies, temporarily compressing the elite advantage. Consistent with the population level findings, Season 5 remains borderline insignificant, though decks in the top 100 measured a marginal increase of hero advantage compared to decks in the top 1,000.

Having established the statistical aspect of hero dominance at both population and elite levels, the following section examines how these competitive dynamics were reflected in the sentiment expressed by active competitive players throughout the hero era.

6 Sentiment Analysis

Statistical advantages of in-game mechanics, as discussed in Section 5, are not the only contributor to what makes players believe a game is overly monetized. Player sentiment, especially with regards to the general feelings of unfairness, provide meaningful signals of balancing issues within intensely competitive games like Clash Royale. After collecting 3,499 tweets from 11 active top players within the Clash Royale community, VADER sentiment analysis was utilized to discern player sentiment toward the state of the game. Tweets were split into the same date ranges of the pre-determined 5 seasons, with Figure 4 numerically displaying the relative sentiment score of each season. As seen in the graph on the top left, players viewed the game the most positive in Season 4, and the most negative in Season 5.

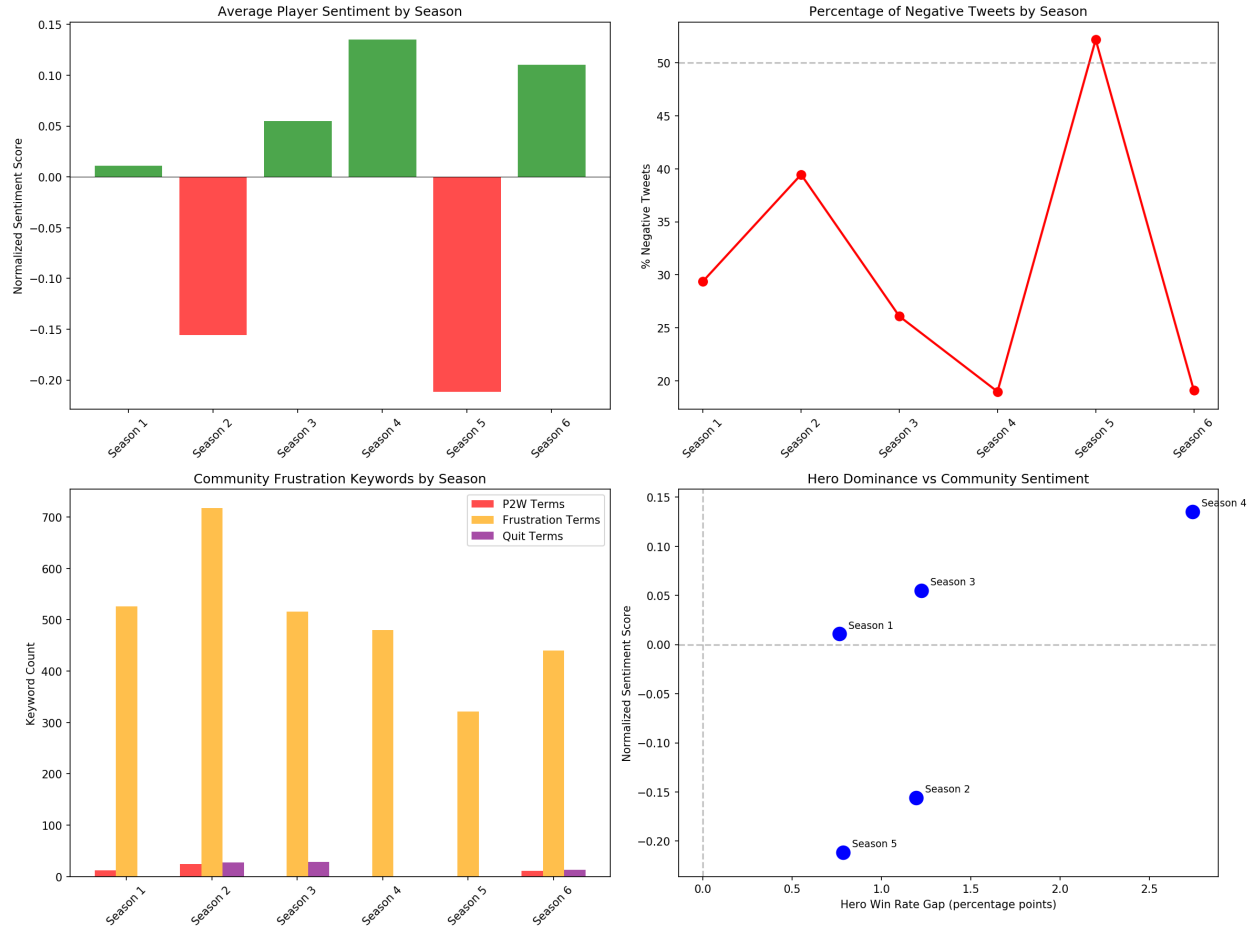


Figure 4: Seasonal Sentiment Analysis — Average Sentiment, Negative Tweet Percentage, Keyword Frequencies, and Hero Dominance vs Sentiment

By applying a keyword filter on derogatory terms, split into P2W and frustration terms, such as “pay to win”(P2W), “scam”, “braindead”, “unbalanced”, and “money grab”, as well as “quit” terminology representing players reaching their breaking point, VADER indicated that Season 2 consisted of the most negative normalized sentiment of -0.156, leading with 718 documented frustration terms and 27 quit terms, as shown by the graphs on the top and bottom left. The elevated quit term frequency observed in Seasons 2 and 3 coincides with the February 2026 departure of prominent content creator Jynxzi, whose public exit from the game citing hero-related concerns represented the most visible community flashpoint of the study period. Additionally, from the

graph in the top right and bottom right, Season 5 displayed the greatest percentage of negative tweets, despite having a statistically insignificant hero win rate advantage as mentioned before. Season 4, the most statistically significant hero season, paradoxically had the most positive normalized sentiment score, and P2W terms remained sparse, heavily concentrated in Seasons 1 and 2 and largely absent in the later seasons.

While the sentiment data provides directional observations about community mood throughout the hero era, its interpretation is complicated by the limited sample of 11 accounts and the video-first communication style prevalent among competitive players, which likely suppresses measurable negative sentiment. The disparity between the statistical dominance observed in Section 5 and the non-linear sentiment patterns presented here raises substantive questions about the relationship between competitive advantage and perceived fairness which are explored in depth in the following section.

7 Discussion¹

From the findings of Section 5, heroes exhibited statistical dominance as decks including them thrived, while decks without them seemed to be lacking. It is worth distinguishing statistical from practical significance when interpreting these results. With sample sizes of 421,436 games at the population level and 143,568 games among elite players, even modest win rate gaps register as statistically significant at conventional thresholds. The Season 1, Season 2, and elite Season 2 gaps of 0.77%, 1.19%, and 1.08% respectively, while statistically distinguishable from zero, represent a meaningfully smaller competitive edge than the 2.74% to 3.68% gaps observed in Seasons 3 and 4. Statistical significance in this context confirms that hero decks reliably outperformed non-hero decks rather than the reverse; the magnitude of that outperformance, and its practical relevance to a player's odds of winning, varied considerably across seasons and is better assessed from the win rate gap and its confidence interval (Appendix B, Tables 2–3) than from the p-value alone.

¹The author's active participation as a top 1,000 Clash Royale ladder player informed qualitative interpretations throughout this section.

This remained a consistent feature across accomplished players in the top 1,000, reinforced by an even stronger advantage observed among elite players in the top 100. The amplification of hero advantage at elite level is consistent with heroes providing mechanical returns that scale with player skill, suggesting genuine competitive superiority rather than a mere perception effect. This interpretation carries an important caveat: hero adoption is not randomly assigned. Players choose whether to run a hero, and that choice may correlate with other factors such as deck archetype quality, meta-awareness, or time investment, that independently predict win rate. The observed gap therefore reflects an association between hero presence and performance rather than a fully isolated causal effect of the hero mechanic itself. The competitive nature of Clash Royale incentivizes top players to take any competitive advantage they had access to, and the strength of heroes since their release has only incentivized more top players to use them. The use of heroes, paired with raw skill within the game, culminated in a remarkable 95% hero adoption rate among top 100 finishers in Season 4. While heroes represent one dimension of Clash Royale's broader monetization ecosystem, their incorporation through exclusive microtransactions provides a measurable and statistically verifiable benchmark for studying the relationship between spending and competitive advantage.

Season 4's paradoxical combination of peak statistical hero dominance and positive community sentiment may be partially explained by concurrent changes to hero accessibility. The March 2026 update introduced a hero album event allowing players to unlock Magic Archer, the season's most statistically dominant hero, through active gameplay over approximately three weeks, requiring four daily battles to complete. This represented a meaningful departure from the purely luck or purchase based acquisition model of earlier seasons, and players appeared to respond positively to Supercell's apparent responsiveness to accessibility concerns. However, this optimism warrants scrutiny as the Barbarian Barrel, the season's second hero release, remained exclusively accessible through paid means, and the Magic Archer unlock demanded sustained daily engagement over an extended period. The positive sentiment of Season 4 may therefore reflect community appreciation for incremental progress rather than genuine satisfaction with the hero system's overall fairness,

suggesting that perceived developer responsiveness can decouple community sentiment from underlying competitive dynamics.

The elevated negative sentiment observed in Seasons 2 and 5 may reflect factors beyond hero dominance specifically. Season 2 coincided with a graveyard rework widely perceived as severely unbalanced, prompting an emergency nerf from Supercell within a week of release: a rare intervention that signals the magnitude of community frustration during that period. Similarly, Season 5's elevated negativity likely reflects anticipation of the Bowler hero teaser released near the season's end, which generated significant community concern over its apparent ability to attack troops across the arena at high damage output. These confounding events suggest that while sentiment data captures general community mood accurately, attributing seasonal negativity exclusively to hero dynamics would be an oversimplification, a limitation acknowledged when interpreting the relationship between statistical hero dominance and measured sentiment.

When discussing monetization, all is relative. The sought-after balance between player satisfaction and company revenue varies from mobile game to mobile game. For a game like Clash Royale where game fundamentals remain relatively constant, increased monetization often becomes correlated with content updates, as additions of new cards and mechanics sometimes go hand in hand with pre-release offers and new currency systems. Therefore, the broad "pay to win" debate often reflects the multifaceted nature of monetization, serving more as a spectrum rather than a binary choice between monetization and player satisfaction. Community perception of what is "fair" and what is not also varies, playing a large role in how tolerant players are of any changes implemented. Further investigation of mobile game monetization would benefit from incorporating video-based sentiment analysis tools capable of capturing frustration expressed through gameplay clips: a communication style particularly prevalent in competitive gaming communities. Expanding the player sample and extending the time horizon beyond the hero era's initial seasons would additionally strengthen the generalizability of findings presented here.

8 Conclusion

The addition of unique in-game mechanics such as heroes in Clash Royale illuminates the monetization nuances inherent to competitive mobile gaming. In four of the five investigated seasons, hero decks were statistically dominant over non-hero decks, a trend further amplified among elite players in the top 100. Season 4 marked the peak of this dominance, with a 95% hero deck adoption rate among top 100 finishers and Magic Archer emerging as the season’s most dominant individual hero. Season 5 subsequently displayed a waning of hero advantage, perhaps suggesting a gradual return to competitive normalcy.

While quantitative measures of hero advantage consistently increased before declining in Season 5, player sentiment provided a complementary perspective from top players. Despite relative hero strength in Season 2, negative keywords and frustration terms peaked that season, as compared to Season 4, which saw paradoxically positive comments on the state of the game, though the enabling of players to unlock heroes for free may have meaningfully shaped community sentiment toward the hero system. In-game mechanics dictate how the game itself flows, but player emotions play just as large a role in finding balance in monetization, exemplifying the intricacies mobile game developers have to navigate when considering new features.

Due to battle data collection ceasing in May 2026, Season 6 could not be incorporated into the statistical analysis, precluding observation of the hero era’s more recent competitive dynamics. Top 100 leaderboard data was manually compiled from RoyaleAPI blog posts and while no systematic errors were identified during review, the potential for minor transcription errors exists. Additionally, due to platform constraints, sentiment analysis was limited to 11 successfully retrieved Twitter/X accounts, a sample size that restricts generalizability. VADER’s known limitations in interpreting sarcasm and gaming-specific vernacular introduce further measurement uncertainty into the sentiment component. Additionally, the game-level t-test approach treats each game as an independent observation while individual players contribute multiple games within the two-day window. A player-level aggregation was explored but produced results difficult to interpret cleanly due to the shrinking non-hero player pool in later seasons, and non-independence is

therefore flagged as a limitation rather than a resolved concern.

As the hero era continues to evolve beyond the study period, future work extending this methodology to additional seasons and broader player samples would further illuminate the long-term relationship between hero accessibility, competitive dominance, and community perception in Clash Royale and mobile gaming more broadly.

References

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- [5] Hutto, C. J., & Gilbert, E. (2014). VADER: A parsimonious rule-based model for sentiment analysis of social media text. *Proceedings of the Eighth International Conference on Weblogs and Social Media*. <https://ojs.aaai.org/index.php/ICWSM/article/view/14550>

A Hero Card Reference Table

Card Name	Card ID	Release Date	Season Introduced
Knight	26000000	December 1, 2025	Season 1
Musketeer	26000014	December 1, 2025	Season 1
Giant	26000003	December 1, 2025	Season 1
Mini PEKKA	26000018	December 1, 2025	Season 1
Wizard	26000017	January 5, 2026	Season 2
Ice Golem	26000038	January 16, 2026	Season 2
Goblins	26000002	February 2, 2026	Season 3
Mega Minion	26000039	February 13, 2026	Season 3
Magic Archer	26000062	March 2, 2026	Season 4
Barbarian Barrel	28000015	March 2, 2026	Season 4
Balloon	26000006	April 4, 2026	Season 5
Dark Prince	26000027	May 4, 2026	Season 6
Bowler	26000034	May 4, 2026	Season 6

Table 1: Hero Card Reference Table

B Full Seasonal Results

Season	Hero WR	95% CI	Non-Hero WR	95% CI	P-Value
Season 1	50.38%	[49.89%, 50.87%]	49.62%	[49.13%, 50.11%]	0.0304
Season 2	50.60%	[50.14%, 51.05%]	49.40%	[48.95%, 49.86%]	0.0003
Season 3	50.61%	[50.00%, 51.22%]	49.39%	[48.78%, 49.99%]	0.0055
Season 4	51.37%	[50.67%, 52.07%]	48.63%	[47.93%, 49.33%]	<0.0001
Season 5	50.39%	[49.81%, 50.98%]	49.61%	[49.02%, 50.19%]	0.0611

Table 2: Population Level Win Rate Results with 95% Confidence Intervals (Top 1,000)

Season	Hero WR	95% CI	Non-Hero WR	95% CI	P-Value	Top 100 Adoption
Season 1	51.42%	[50.60%, 52.25%]	48.58%	[47.76%, 49.40%]	<0.0001	60%
Season 2	50.54%	[49.79%, 51.29%]	49.46%	[48.71%, 50.21%]	0.0454	67%
Season 3	51.84%	[50.67%, 53.01%]	48.16%	[46.99%, 49.33%]	<0.0001	82%
Season 4	51.72%	[50.50%, 52.94%]	48.28%	[47.06%, 49.50%]	<0.0001	95%
Season 5	50.67%	[49.60%, 51.74%]	49.33%	[48.26%, 50.40%]	0.0832	83%

Table 3: Elite Level Win Rate Results with 95% Confidence Intervals (Top 100)

C Sentiment Keyword Lists

Category	Keywords
P2W Terms	p2w, pay to win, pay2win, cash grab, scam, predatory, greedy, money grab
Frustration Terms	broken, op, nerf, unfair, ridiculous, braindead, joke, disgusting, trash, garbage, unbalanced, stupid, pathetic, awful, terrible, unplayable, cancer, toxic, clown, fraudulent
Quit Terms	quit, quitting, uninstall, deleted, done with, leaving, left the game, jynxzi

Table 4: Sentiment Analysis Keyword Lists